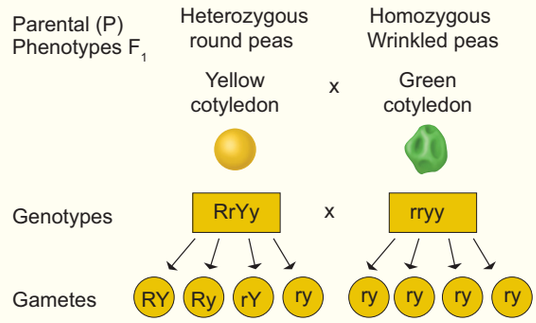
### 2.3.5 The Dihybrid test cross

**The Dihybrid test cross**

The $F_1$ hybrid plant (round yellow peas) is crossed with homozygous double recessive genotype, wrinkled green peas (rryy). This is called dihybrid test cross with the ratio of 1:1:1:1.




| Gametes | RY | Ry | rY | ry |
|---|---|---|---|---|
| all ry | RrYy Round Yellow | Rryy Round Green | rrYy Wrinkled Yellow | rryy Wrinkled Green |

| Genotypes | RrYy | Rryy | rrYy | rryy |
|---|---|---|---|---|
| Genotypic ratio | 1 : | 1 : | 1 : | 1 |
| Phenotypes | round Yellow | round Green | Wrinkled Yellow | Wrinkled Green |
| Phenotypic ratio | 25% | 25% | 25% | 25% |

**Figure 2.12:** Dihybrid test cross

### 2.3.6 Extensions of Mendelian Genetics

Apart from monohybrid, dihybrid and trihybrid crosses, there are exceptions to Mendelian principles, i.e. the occurrence of different phenotypic ratios. The more complex patterns of inheritance are the extensions of Mendelian Genetics. There are examples where phenotype of the organism is the result of the interactions among genes.

## 2.4 Gene interaction

A single phenotype is controlled by more than one set of genes, each of which has two or more alleles. This phenomenon is called Gene Interaction. Many characteristics of the organism including structural and chemical which constitute the phenotype are the result of interaction between two or more genes.

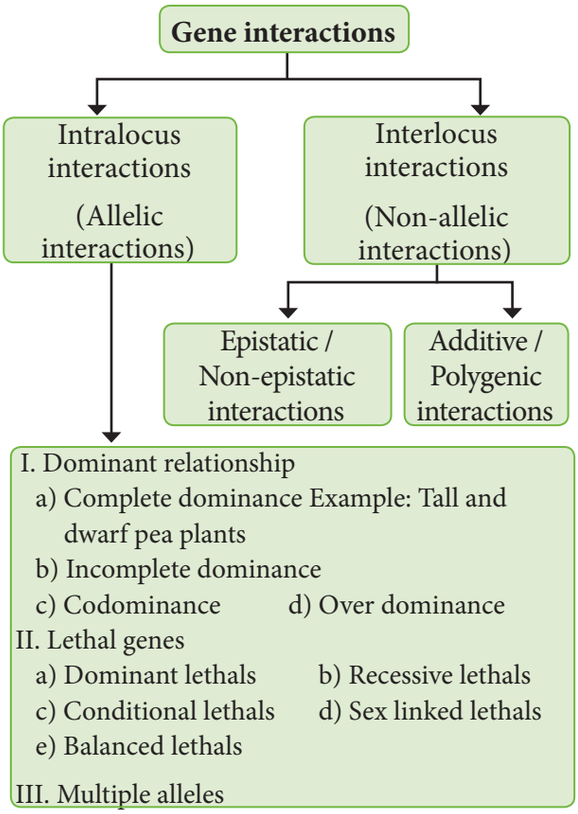



**Figure 2.13:** Gene Interaction

Mendelian experiments prove that a single gene controls one character. But in the post Mendelian findings, various exception have been noticed, in which different types of interactions are possible between the genes. This gene interaction concept was introduced and explained by W. Bateson. This concept is otherwise known as Factor hypothesis or Bateson's factor hypothesis. According to Bateson's factor hypothesis, the gene interactions can be classified as

- Intragenic gene interactions or Intra allelic or allelic interactions
- Intergenic gene interactions or inter allelic or non-allelic interactions

### 2.4.1 Intragenic interactions

Interactions take place between the alleles of the same gene i.e., alleles at the same locus is called intragenic or intralocus gene interaction. It includes the following:
a) Incomplete dominance (b) Codominance (c) Lethal genes (d) Pleiotropic genes are common examples for intragenic interaction.



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